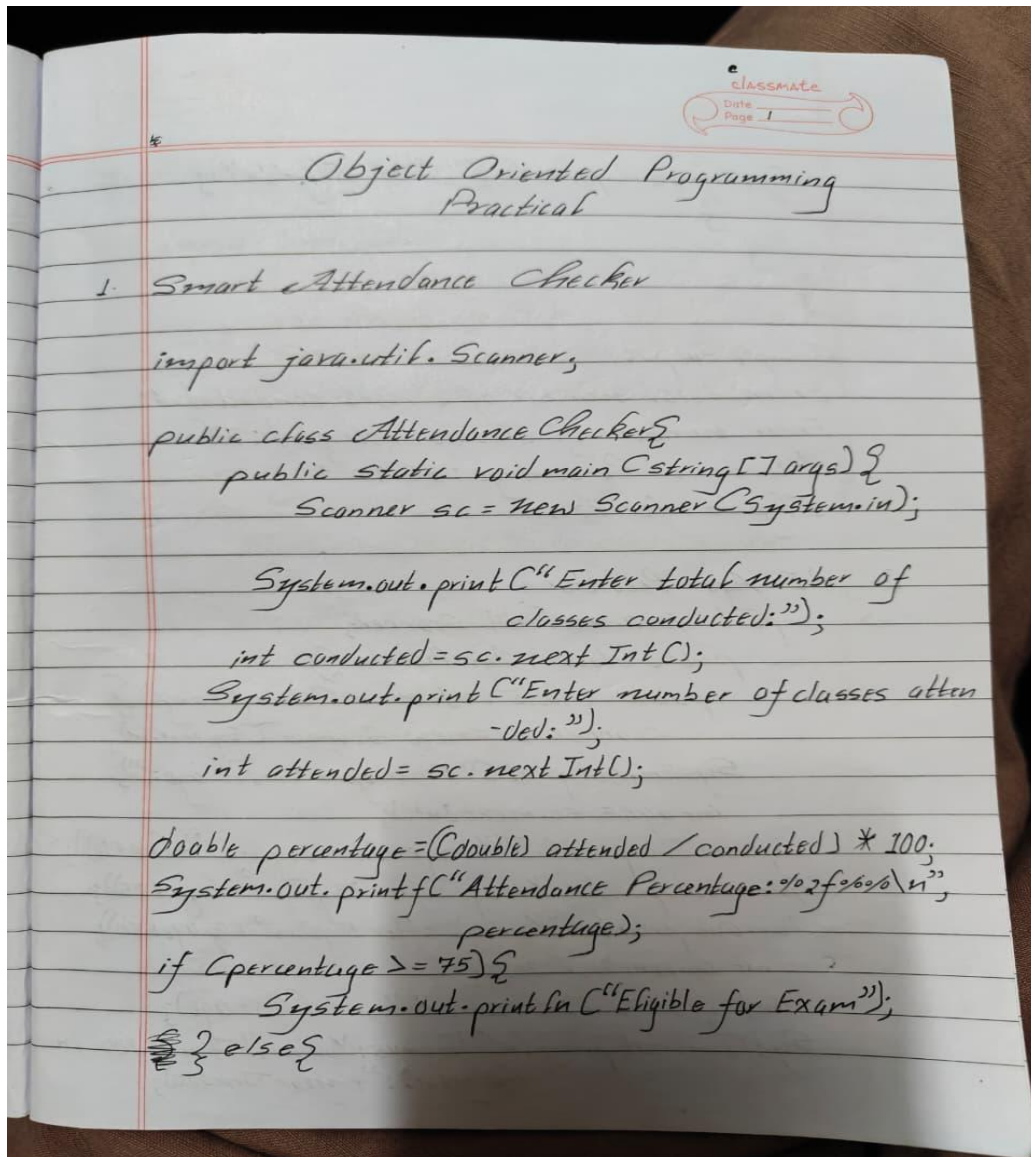


# OOP Assignment

## 1. Smart Attendance Checker



```
"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=5290
Enter total number of classes conducted: 60
Enter number of classes attended: 70
Attendance Percentage: 116.67%
Eligible for Exam

Process finished with exit code 0
```

## 2. AI Age Predictor

```
System.out.println("Not Eligible");  
    }  
    }  
}
```

Sample output:

Enter total number of classes conducted: 40  
Enter number of classes attended: 32  
Attendance Percentage: 80.00 %  
Eligible for Exam.

### 2. AI Age Predictor

```
→ import java.util.*; Scanner;  
public class Age Predictor {  
    public static void main (String[] args) {  
        Scanner sc = new Scanner (System.in);  
        System.out.println("Enter your current age:");  
        int age = sc.nextInt();  
        System.out.println("Your age after 10 years: " + (age + 10));  
        System.out.println("Your age after 25 years: " + (age + 25));  
        System.out.println("Your age after 50 years: " + (age + 50));  
        int currentYear = 2026;  
        int yearTurn100 = currentYear + (100 - age);  
        System.out.println("Bonus: You will turn 100 in  
        the year: " + yearTurn100);  
    }  
}
```

```
"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=5292  
Enter your current age: 19  
Your age after 10 years: 29  
Your age after 25 years: 44  
Your age after 50 years: 69  
Bonus: You will turn 100 in the year: 2107  
  
Process finished with exit code 0
```

### 3. Fitness Challenger Tracker

#### 3. Fitness Challenger Tracker

```
import java.util.Scanner;
public class FitnessTracker {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        int[] steps = new int[7];
        int total = 0;
        int highest = 0;
        for (int i = 0; i < 7; i++) {
            System.out.print ("Enter steps walked on day " +
                               (i+1) + ": ");
            steps[i] = sc.nextInt();
            total += steps[i];
            if (steps[i] > highest) {
                highest = steps[i];
            }
        }
        double average = (double) total / 7;
        System.out.println ("--- Results ---");
        System.out.print ("Total steps: " + total);
        System.out.print ("Average steps: %.2f", average);
        System.out.print ("Highest number of steps: " + highest);
    }
}
```

```
"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=5292
Enter steps walked on day 1: 7000
Enter steps walked on day 2: 9000
Enter steps walked on day 3: 10000
Enter steps walked on day 4: 8000
Enter steps walked on day 5: 5000
Enter steps walked on day 6: 3000
Enter steps walked on day 7: 15000

--- Results ---
Total steps: 57000
Average steps: 8142.86
Highest number of steps: 15000

Process finished with exit code 0
```

## 4. Fibonacci Series

### 4. Fibonacci Series

```
import java.util.Scanner;
public class Fibonacci {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the value of n:");
        int n = sc.nextInt();

        int first = 0, second = 1;
        System.out.print("Fibonacci Series up to "
            + n + " terms:");

        for (int i = 1; i <= n; i++) {
            System.out.print(first + " ");
            int next = first + second;
            first = second;
            second = next;
        }
        System.out.println();
    }
}
```

"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea\_rt.jar=59589" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8

Enter the value of n: 67

Fibonacci Series up to 67 terms: 0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765 10946 17711 28657 46368 75025 121393 196418 317811 514229

Process finished with exit code 0



## 5. Prime Number Check

### 5. Prime Number Check

```
import java.util. Scanner;
public class Prime Check {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter a number: ");
        int num = sc.nextInt();

        boolean isPrime = num > 1;
        for (int i = 2; i <= (Math.sqrt (num)); i++) {
            if (num % i == 0) {
                isPrime = false;
                break;
            }
        }
        if (isPrime) {
            System.out.print (num + " is a prime number.");
        } else {
            System.out.print (num + " is not a prime number.");
        }
    }
}
```

```
"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=59606" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Enter a number: 90
90 is not a prime number.

Process finished with exit code 0
```

## 6. Logical Operators Check

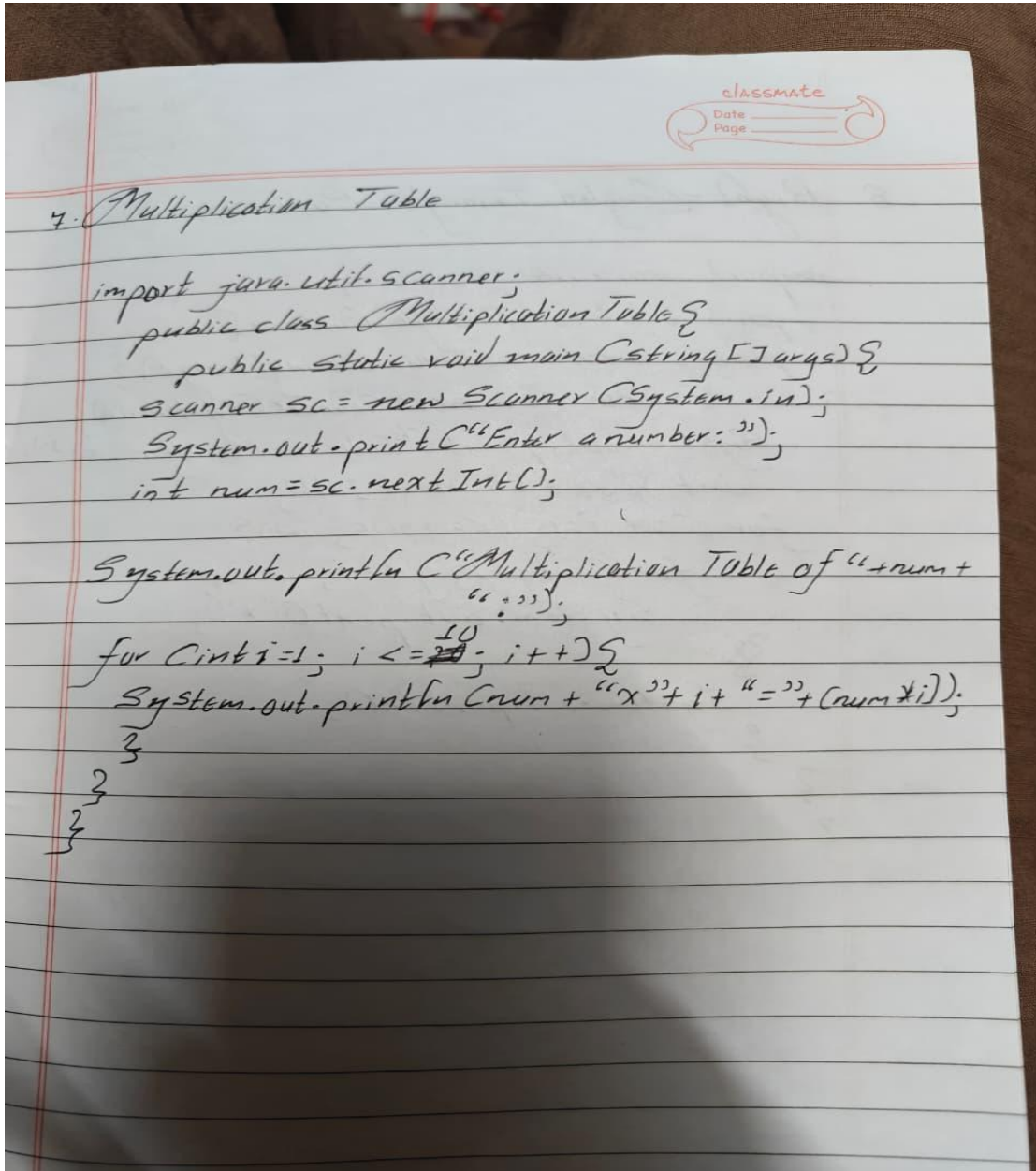
classmate  
Date \_\_\_\_\_  
Page \_\_\_\_\_

6. Logical Operators Check

```
import java.util.Scanner;  
public class LogicalOperatorsCheck {  
    public static void main (String[] args) {  
        Scanner sc = new Scanner (System.in);  
  
        System.out.println("Enter first number:");  
        int num1 = sc.nextInt();  
        System.out.println("Enter second number:");  
        int num2 = sc.nextInt();  
        System.out.println("Enter third number:");  
        int num3 = sc.nextInt();  
  
        boolean check = (num1 + num2 == num3) && (num3 -  
            num2 == num1);  
  
        if (check) {  
            System.out.println("True: the third number is the  
                sum of the first two.");  
        } else {  
            System.out.println("False: the third number is  
                NOT the sum of the first two.");  
        }  
    }  
}
```

```
"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=54822" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8  
Enter first number: 7  
Enter second number: 3  
Enter third number: 1  
False: The third number is NOT the sum of the first two.  
  
Process finished with exit code 0
```

## 7. Multiplication Table



```
"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=54856" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Enter a number: 67
Multiplication Table of 67:
67 x 1 = 67
67 x 2 = 134
67 x 3 = 201
67 x 4 = 268
67 x 5 = 335
67 x 6 = 402
67 x 7 = 469
67 x 8 = 536
67 x 9 = 603
67 x 10 = 670
Process finished with exit code 0
```

## 8. Right-Angled Triangle Pattern

### 8. Right-Angled Triangle Pattern

```
import java.util.Scanner;
public class Triangle Pattern {
    public static void main (String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();
        for (int i=1; i<= rows; i++) {
            for (int j=1; j<= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea\_rt.jar=54918" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8

Enter number of rows: 4

```
*
**
***
****
```

Process finished with exit code 0



## 9. Pyramid Pattern

### 9) Pyramid Pattern

```
import class PyramidPattern {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter number of rows:");
        int rows = sc.nextInt();
        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= rows - i; j++) {
                System.out.print (" ");
            }
            for (int k = 1; k <= (2 * i - 1); k++) {
                System.out.print ("*");
            }
            System.out.println();
        }
    }
}
```

"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea\_rt.jar=51130" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8

Enter number of rows: 5

```
*
**
***
****
*****
*****
*****
```

Process finished with exit code 0

## 10. Palindrome Number

### 10. Palindrome Number

```
public class PalindromeNumber {  
    public static boolean isPalindrome(int x) {  
        if (x < 0) return false;  
        int original = x;  
        int reversed = 0;  
        while (x != 0) {  
            int digit = x % 10;  
            reversed = (reversed * 10) + digit;  
            x /= 10;  
        }  
        return original == reversed;  
    }  
    public static void main (String[] args) {  
        int testNum = 121;  
        System.out.println("Is " + testNum + " a palindrome?  
        - true" + isPalindrome(testNum));  
        testNum = -121;  
        System.out.println("Is " + testNum + " a palindrome?  
        - false" + isPalindrome(testNum));  
    }  
}
```

```
"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=54176" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -[  
Is 121 a palindrome? true  
Is -121 a palindrome? false  
Process finished with exit code 0
```

# 11. Basic Calculator

## 11. Basic Calculator

```
import java.util.Scanner;

public class Basic Calculator {
    public static void main (String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print ("Enter first number: ");
        double n1 = sc.next Double ();
        System.out.print ("Enter an operator (+, -, *, /): ");
        char op = sc.next ().char At (0);
        System.out.print ("Enter second number: ");
        double n2 = sc.next Double ();
        switch (op) {
            case '+': System.out.println ("Result: " + (n1 + n2));
                        break;
            case '-': System.out.println ("Result: " + (n1 - n2)); break;
            case '*': System.out.println ("Result: " + (n1 * n2));
                        break;
            case '/':
                if (n2 != 0) {
                    System.out.println ("Results: " + (n1 / n2));
                } else {
                    System.out.println ("Error: Divison by zero
                    is under defined.");
                }
            }
    }
}
```

```
"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=54182" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Enter first number: 8
Enter an operator (+, -, *, /): -
Enter second number: 9
Result: -1.0

Process finished with exit code 0
```

## 12. Grade Calculator

```
break;  
default: System.out.println("Invalid operator!");  
}  
}  
}
```

### 12. Grade Calculator

```
import java.util.Scanner;  
public class GradeCalculator {  
    Scanner sc = new Scanner(System.in);  
    System.out.print("Enter marks obtained:");  
    double marks = sc.nextDouble();  
  
    String grade = (marks >= 85) ? "A":  
        (marks >= 75) ? "A-":  
        (marks >= 70) ? "B+":  
        (marks >= 65) ? "B":  
        (marks >= 60) ? "B-":  
        (marks >= 55) ? "C+":  
        (marks >= 50) ? "C":  
        (marks >= 45) ? "D": "F";  
    System.out.println("Grade: " + grade);  
}
```

```
"C:\Program Files\Java\latest\jdk-26\bin\java.exe" "-javaagent:D:\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=54190" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF
```

```
Enter marks obtained: 77
```

```
Grade: A-
```

```
Process finished with exit code 0
```